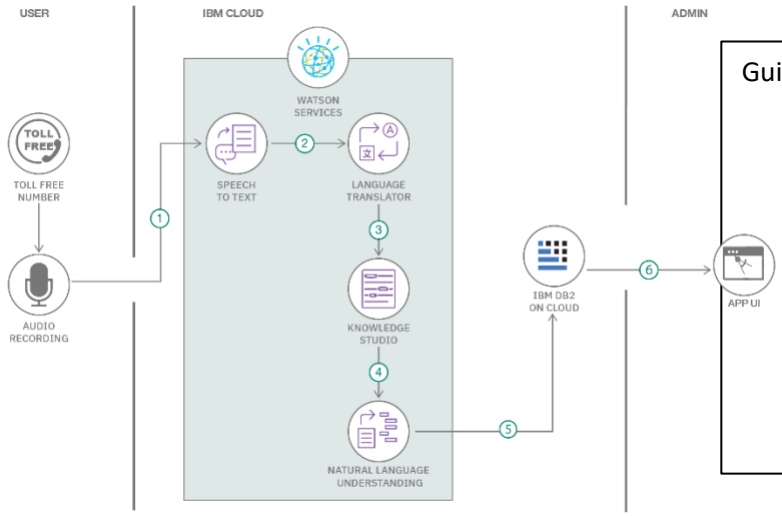
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS90862
Project Name	Prosperity Prognosticator: Machine Learning for Startup Success Prediction
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram of the Startup Success Prediction system as shown below and the information as per Table 1 & Table 2.

Example: Startup Success Prediction using Machine Learning and Flask Web



Guidelines:

Include all the processes (As application logic / Machine Learning and Flask Technology Block)
Provide infrastructural demarcation (Local Flask Server / Cloud Deployment if applicable)
Indicate external interfaces (Python libraries such as Scikit-learn, Pandas, Joblib, etc.)
Indicate Data Storage components / services (CSV dataset files and saved .pkl model file)
Indicate interface to machine learning models (Integration of trained Random Forest model with Flask application)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web interface where users enter startup details and view prediction results	HTML, CSS, JavaScript, Bootstrap
2	Application Logic	Backend logic for handling requests, data validation, and prediction processing	Python (Flask Framework)

3	Data Preprocessing Module	Handles feature selection, scaling, and transformation of input data	Pandas, NumPy, Scikit-learn
4	Machine Learning Model	Predicts startup success or failure using classification algorithm	Random Forest (Scikit-learn)
5	Database	Stores user credentials and prediction logs	MySQL / SQLite
6	Model Storage	Stores trained ML model file	Joblib / Pickle (.pkl file)
7	File Storage	Stores dataset CSV files used for training and testing	Local File System
8	External Dataset Source	Historical startup dataset used for model training	Kaggle Dataset (CSV format)
9	API Integration	Handles prediction request and response between frontend and backend	Flask REST API
10	Infrastructure (Server / Cloud)	Application deployment environment	Local Server / Cloud (if deployed)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Use of open-source tools and libraries for development	Flask, Scikit-learn, Pandas
2	Security Implementations	Secure login, password hashing, and session management	Flask Login, SHA-256 hashing
3	Scalable Architecture	Modular structure separating frontend, backend, and ML model	3-Tier Architecture (UI, Logic, Model)
4	Availability	Application accessible through web interface	Localhost / Web Server
5	Performance	Fast prediction response within few seconds	Optimized Random Forest model